TIMBRE MORPHING VIA WEIGHT INTERPOLATION OF SPECIALIZED VARIATIONAL AUTOENCODERS

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ABSTRACT

Timbre morphing, the generation of intermediate timbres between known sounds, expands the sonic palette for music production and sound design. Common techniques interpolate in the latent space of a single generative model trained on multiple timbres, which in small models compromises the fidelity of each one. We propose an alternative: training variational autoencoders (β-VAE) specialized per timbre, and generating hybrid timbres by interpolating directly in the model weight space. The system is trained on STFT magnitude spectrograms from four sources (piano, guitar, voice and bass) and evaluated with Fréchet Audio Distance over MERT embeddings. A 2×2 ablation shows that fine-tuning every model from a common base model, rather than training it from scratch, makes the transfer towards the target monotonic, while latent-space regularization keeps unrelated timbres unaffected; only their combination yields gradual and predictable transitions. The models are small, and the weight-interpolated decoder runs in real time in a web browser.

1. INTRODUCTION

Timbre is the perceptual attribute that lets us distinguish two sounds of equal pitch and loudness produced by different sources. Generating intermediate timbres between two known sounds—timbre *morphing*—is of particular interest for music production and sound design, as it enables hybrid sonorities that belong to no single source.

The most widespread deep-learning approaches learn a single generative model over multiple timbres and interpolate within its latent space; however, when the model is small this strategy tends to compromise the fidelity of each individual timbre (in our setting, per-instrument models reconstruct better than a shared model of the same size; Sec. 4.1). This work explores an alternative: training a specialized model per timbre and interpolating directly in the space of their *weights*, so as to preserve the reconstruction quality of each source model. A design principle throughout is to keep the models deliberately small, so that both inference and weight interpolation run on modest hardware and in real time, aiming at a practical tool for performance and production.

Contributions. (i) We propose a timbre-morphing technique that interpolates the *weights* of per-instrument specialized VAEs, rather than the latent code of a single shared model, and evaluate it on four sound sources. (ii) Through a 2×2 ablation over initialization (shared checkpoint vs. scratch) and latent regularization (β-VAE vs. AE), we provide initial evidence that both factors contribute to well-behaved interpolations, each to a different aspect. (iii) We show that the weight-interpolated decoder runs in real time in a web browser, and release code, models, audio examples and an interactive demo. [1]

2. RELATED WORK

Audio morphing can be organized by *where* the interpolation takes place: on the **signal**, on the **latent space** of a generative model, or—as we propose—on the model **weights**. A foundational signal-domain method [1] aligns two spectrograms with dynamic time warping and cross-fades them; it is training-free but requires explicit alignment and was validated only on voice. With deep learning, morphing moved to the latent space. NSynth [2] interpolates note embeddings in a *single* WaveNet-decoder autoencoder trained on all instruments, and Magenta turned it into an interactive grid [3] close to the tool motivating our work. Its autoregressive decoder, however, precludes real-time synthesis.

Recent work relies on latent diffusion, with high quality at high cost. Latent diffusion bridges [4] train one diffusion model per instrument connected by a shared prior, but perform full timbre *transfer* rather than controllable intermediate states. SoundMorpher [5] morphs in the noise and embedding space of a pretrained text-to-audio model and introduces perceptually uniform morphing, at the cost of a large model. For guitar tones, Chen *et al.* [6] find that a lightweight autoencoder beats fine-tuned diffusion, and one variant interpolates network weights, though restricted to a single instrument. Mix2Morph [7] trains a large text-to-audio model to morph from noisy mixes. Overall, the literature interpolates the *content* inside a fixed model.

[1] https://github.com/facundo2gonzalez2/2ASMRS

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Interpolating network *weights* has been studied mainly outside audio. Averaging models fine-tuned from a shared pre-trained checkpoint improves accuracy at no inference cost [8], and weight differences act as composable "task vectors" [9]. This works because models trained from a common starting point stay in the same loss basin and are linearly connected [10, 11], which motivates our *Checkpoint* strategy. Closest to our setting, Murata *et al.* [12] merge text-to-speech models to interpolate speaker and emotion attributes. Spherical interpolation [13] is common for latent vectors of generative models [14]; we apply it per parameter tensor.

Variational autoencoders (VAE) [15] impose a probabilistic structure on the latent space via the Kullback–Leibler (KL) divergence, which facilitates sampling and interpolation; the β-VAE [16] weights this term by β. For music sound modeling, variational models offer the best fidelity/regularization trade-off [17]; RAVE [18] showed real-time VAE synthesis; and sparse autoencoders build perceptually consistent timbre spaces [19].

3. METHOD

3.1 Dataset and Preprocessing

We use four instruments with clearly distinguishable timbres: piano, guitar, bass and voice, with about 87 minutes of audio each (5.8 hours in total, balanced to avoid biasing training). Piano comes from 20 freely licensed tracks: Mozart performances by M. Staab and B. Kinsella (via Chosic), royalty-free tracks from Pixabay, and nine AI-generated solo-piano samples from ElevenLabs (per-track sources and licenses are listed in the repository); guitar from a subset of GuitarSet [20] (electric guitar, melodic and chordal passages); bass from jazz-standard basslines in FiloBass [21]; and voice from a subset of VocalSet [22] (two male and two female voices, with melodies, scales and arpeggios).

All tracks are converted to mono at 22050 Hz and analyzed with an STFT (2048-sample Hann window, 512-sample hop). The model works only with the magnitude, in decibels $S = 10\log_{10}(|F|^2)$; values below -60 dB are clipped, the floor is shifted to zero and the result is divided by the global maximum X_{max} (stored with the model), mapping magnitudes to $[0, 1]$. Each 1025-bin frame is an independent input to the network.

3.2 Variational Autoencoder and Architecture

The encoder approximates the posterior $q(z \mid x)$ and the decoder the likelihood $p(x \mid z)$, with prior $p(z) = \mathcal{N}(0, I)$. The β-VAE [16] minimizes

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \beta\,\mathcal{L}_{\text{KL}}, \tag{1}$$

where $\mathcal{L}_{\text{rec}}$ is the mean squared error (MSE) between input and reconstruction and $\mathcal{L}_{\text{KL}}$ the closed-form KL divergence to the prior; z is sampled with the reparameterization trick. When $\beta = 0$ the model reduces to a deterministic autoencoder (AE) and we take $z = \mu$. Higher β yields a smoother latent space at the cost of fidelity.

The encoder is a stack of fully connected layers $1025 \to (1024, 512, 256, 128, 64, 32, 16, 8) \to 4$, each hidden layer combining a linear map, Batch Normalization (BN) and an ELU activation, with two heads for μ and $\log\sigma^2$. The decoder mirrors it back to 1025 bins, with a final linear layer and BN but no activation. The full model has 3.5 M parameters, of which 1.76 M belong to the decoder (≈ 3.5 MFLOP per frame).

3.3 Training Strategy

We optimize Eq. (1) with Adam (learning rate 10^{-3}), a $95/5$ train/validation split, batches of 256 frames, early stopping on the validation loss and gradient clipping. Per-instrument models follow two initialization strategies. *Scratch*: each model is trained from a random initialization on its instrument only. *Checkpoint*: we first train a *base model* on all four instruments and then fine-tune one model per instrument from that common starting point. Crossing both initializations with two regularization settings—AE ($\beta = 0$) and β-VAE ($\beta = 0.001$)—gives a 2×2 ablation whose baseline, Scratch-AE, uses neither factor.

3.4 Weight Interpolation

Given two models with identical architecture, a source A and a target B, and $\alpha \in [0, 1]$, we build an intermediate model by interpolating *every* parameter tensor of A and B (encoder and decoder) independently; synthesis then uses only the resulting decoder. By construction $\alpha = 0$ reproduces A and $\alpha = 1$ reproduces B. We consider linear interpolation (LERP), $(1-\alpha)\,w_A + \alpha\,w_B$, and spherical interpolation (SLERP), which traverses the arc joining both flattened tensors:

$$\text{SLERP}(w_A, w_B, \alpha) = \frac{\sin\big((1-\alpha)\theta\big)}{\sin\theta}\,w_A + \frac{\sin\big(\alpha\theta\big)}{\sin\theta}\,w_B, \tag{2}$$

where θ is the angle between w_A and w_B, computed per tensor. When $|\cos\theta| > 0.9995$, SLERP is numerically unstable and we fall back to LERP. The BN running means and variances are floating-point and are interpolated like any other tensor. Only the integer BN step counter, which differs between models and is unused at inference, takes the value of the closer model.

3.5 Audio Reconstruction and Metrics

The decoder output is rescaled by X_{max}, converted to linear magnitude and inverted with a phase estimate: Griffin–Lim, PGHI [23] or random phase. For evaluation we use (i) validation MSE, and (ii) a Fréchet Audio Distance (FAD) [24] over embeddings from MERT-v1-95M [25], mapped to a bounded similarity $\text{sim} = \exp(-\text{FAD}/k)$ with $k = 585$ (a monotone rescaling that does not alter any comparison). FAD is computed at the *clip level*, between the distributions of the 768-d MERT frame embeddings (75 frames/s) of two individual clips. In the interpolation experiments each clip has 48 STFT frames (≈ 1.1 s, ≈ 83 MERT frames), so the covariances are rank-deficient and are regularized with εI ($\varepsilon \le 10^{-1}$). Clip-level FAD is

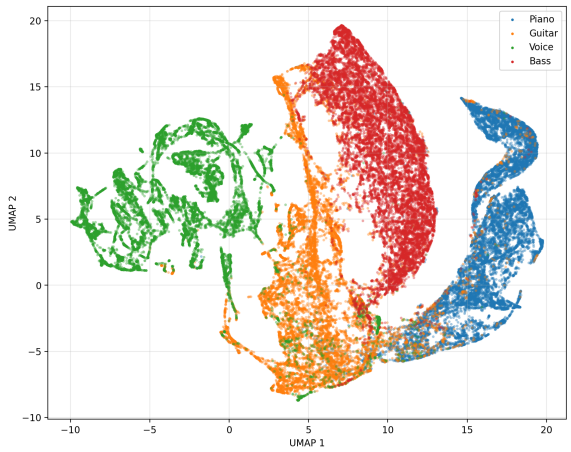


Figure 1: UMAP projection of the latent embeddings of the base model ($\beta = 0.001$), one point per STFT frame, colored by instrument.

known to be biased at small sample sizes [26], so we first check its behaviour on a controlled case (Sec. 4.3).

4. EXPERIMENTS AND RESULTS

4.1 Architecture and β Selection

Sweeping latent dimensions $\{2, 3, 4, 6, 8\}$, a dimension of 4 gives a significant reduction of validation error over 2 and 3, while 6 or 8 add little, consistently for every per-instrument model, for the base model and for the KL term; since four control knobs are far more manipulable than eight, we adopt 4. Sweeping β on the base model, the validation MSE ($\times 10^{-3}$) and KL are $7.71/0.30$ for $\beta = 10^{-2}$, $3.50/1.50$ for 10^{-3}, $2.88/2.85$ for 10^{-4}, $2.28/4.93$ for 10^{-5} and $2.27/386$ for $\beta = 0$. Regularizing more than 10^{-3} doubles the error, while relaxing it lowers the error by 18–35% but doubles or triples the KL; we adopt $\beta = 0.001$ as the compromise that favours a regular latent space, the property interpolation relies on. With the same architecture, the specialized models reconstruct their instrument better than the shared base model: mean MSE 2.06 (both AE configurations) vs. 2.28 for $\beta = 0$, and 3.34 (Checkpoint) vs. 3.61 for $\beta = 0.001$; per-configuration values are in Table 1. Each model is validated on its own split, so this comparison is indicative.

4.2 Latent-Space Analysis

Projecting the base-model latent embeddings to 2-D with UMAP ($n_neighbors = 15$), the four instruments form clearly separated clusters under both $\beta = 0$ and $\beta = 0.001$, with pairwise multidimensional Wilcoxon–Mann–Whitney U statistics close to 1 (0.5 = indistinguishable, 1 = fully separated). Separation is consistently slightly lower with $\beta = 0.001$, as expected from a smoother latent space (e.g. $0.926 \rightarrow 0.796$ for bass–voice; Fig. 1).

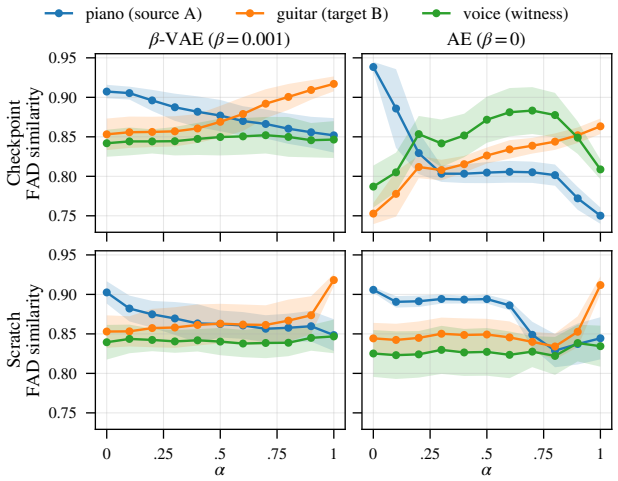


Figure 2: Cross interpolation from piano (A) to guitar (B), with voice as witness: FAD similarity (mean $\pm$ std over 25 latent samples) to each instrument as α sweeps from 0 to 1. Rows: initialization; columns: regularization. Only Checkpoint-β-VAE shows a steady crossover near $\alpha = 0.5$ with a flat witness. Checkpoint-AE transfers abruptly and the witness drifts, and in the Scratch models similarity to B barely grows until $\alpha \approx 0.9$.

4.3 Metric Sanity Check

Before evaluating the models, we check the metric on a case whose outcome is known in advance and that involves no model: a waveform cross-fade $(1 - \alpha)A + \alpha B$ between original clips of two instruments (piano→guitar and voice→piano, 20 random 1.5 s clip pairs each), compared against A, B and a clip of a third, *witness*, instrument. Similarity to A decreases monotonically from 1 to 0.82, similarity to B mirrors it, the curves cross at $\alpha \approx 0.5$, and similarity to the witness stays flat (0.82 ± 0.02). Two clips of different instruments thus score ≈ 0.82 (FAD ≈ 116), which sets the useful range of the metric in our setting; the values in Fig. 2 are similarities, not raw FAD. Cosine similarity between mean MERT embeddings follows the same trends but is more dispersed on time-varying latent inputs, so we use FAD in what follows.

4.4 Weight-Interpolation Experiments

Averaged over every source/target instrument pair, the similarity to the target grows as α increases in all four configurations. Checkpoint-β-VAE shows the smoothest growth and Scratch-AE the worst, delaying the change to extreme values of α.

A stricter test is the cross-interpolation property: as the output approaches B it must move away from A while *not* acquiring traits of a witness instrument that takes no part in the interpolation. Each of 25 latent vectors $z \sim \mathcal{N}(0, I)$ is held constant over a clip and decoded with random phase by the interpolated model and, as references, by the pure A, B and witness models. Fig. 2 shows piano→guitar with voice as witness, and Table 1 summarizes this pair and piano→voice (guitar as witness). Similarity at $\alpha = 0$

Config.	MSE	ρ_A	ρ_B	Δ_W	$\alpha_\times$
Ckpt-β-VAE	3.34	$-.88/-.85$	$.89/.84$	$.03/.04$	$.55/.59$
Ckpt-AE	2.08	$-.81/-.84$	$.95/.90$	$.10/.08$	$.28/.26$
Scratch-β-VAE	3.57	$-.67/-.74$	$.50/.49$	$.04/.05$	$.48/.91$
Scratch-AE	2.04	$-.79/-.58$	$.36/.42$	$.04/.05$	$.76/.40$

Table 1: Per-configuration results. MSE: mean validation reconstruction error over the four instruments ($\times 10^{-3}$). Remaining columns, for piano→guitar / piano→voice over 25 latent samples: Spearman correlation between α and the similarity to A (ρ_A) and to B (ρ_B); range of the similarity to the witness along α (Δ_W); crossover of the mean A and B curves ($\alpha_\times$). Per-sample std of ρ_B is 0.05–0.18 for Checkpoint and 0.26–0.36 for Scratch.

peaks at ≈ 0.91 rather than 1 because each random-phase rendering differs.

The two factors of the ablation affect different aspects of the transition. *Initialization* determines whether the transfer towards B is monotonic: ρ_B is 0.84–0.95 for Checkpoint models but only 0.36–0.50 for Scratch models, whose similarity to B barely grows until $\alpha \approx 0.9$. *Regularization* matters within the Checkpoint models. Without it the transition happens abruptly (crossover at $\alpha \approx 0.27$) and the witness similarity drifts three times more ($\Delta_W = 0.08$–0.10 vs. 0.03–0.04), meaning the intermediate decoders visit timbres unrelated to either endpoint. Only Checkpoint-β-VAE combines a monotonic transfer, a centered crossover ($\alpha_\times = 0.55$–0.59) and a flat witness, consistent with linear connectivity of models fine-tuned from a common checkpoint [10, 11]. Audio examples of continuous transitions for the best and worst configurations, and of audio-to-audio morphing between real recordings, are linked in the repository.

4.5 Real-Time Operation

We implemented the interpolated decoder in a browser synthesizer (ONNX Runtime Web, single-threaded WebAssembly), measured in Firefox on a laptop CPU (Intel Core i7-1255U). SLERP angles depend only on the weights and are precomputed once per model pair. Each frame then evaluates both endpoint decoders ($\approx$ 7 MFLOP) and takes $\approx$ 1–2 ms, and real-time phase reconstruction (RTPGHI) takes 0.6 ms, against a 23.2 ms hop budget. The resulting end-to-end latency is ≈ 70 ms, while α and z are modulated continuously.

5. CONCLUSIONS

We presented a technique to synthesize intermediate timbres by interpolating the weights of per-instrument specialized VAEs, instead of interpolating the latent space of a single model trained on all instruments. A 2×2 ablation shows that interpolation quality depends strongly on the training configuration. Fine-tuning from a common base model makes the transfer monotonic, and latent-space regularization keeps it gradual and free of unrelated timbres. Their combination (Checkpoint-β-VAE) produced the most predictable transitions. The models are small enough for the interpolated decoder to run in real time in a web browser.

Limitations. The evaluation covers four sources and relies on a clip-level FAD over a single embedding model; no formal listening test was conducted, and similarity to the endpoints cannot distinguish a superposition of two sources from a fused timbre. We did not compare against large latent-space morphing systems, whose scale and control paradigm differ. Future work includes a perceptual study, KAD [26] and alternative embeddings, a latent-interpolation baseline on the shared base model, more instruments, and a triple interpolation controlled by two parameters.

6. ETHICS STATEMENT

This work uses publicly available datasets (GuitarSet, FiloBass, VocalSet) under their respective licenses, together with freely licensed piano tracks (some of them AI-generated; see the repository for credits), and does not involve the collection of personal or private data. The proposed generative technique is intended as a creative tool for music production and performance; as with any audio-synthesis method, care should be taken to respect the rights of performers whose recordings are used to train the models.

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